# Laser Beam Profiling & Safety Analysis (KVANT Atom 42 – Simulation-Based Study)

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**Abstract**

High-power solid-state laser systems are widely used in outdoor entertainment and architectural installations. This report presents a simulation-based analysis of Gaussian beam propagation using parameters representative of the KVANT Atom 42 laser system. The study evaluates beam expansion, irradiance variation, wavelength-dependent behavior (RGB), and laser safety considerations. A simplified IEC-based classification approach is also implemented. The results highlight how beam divergence and propagation distance reduce irradiance to safer levels, supporting practical safety measures such as beam scanning and exclusion zones.

## 1 Introduction

High-power laser systems used in entertainment applications operate in open environments where human exposure is possible. Understanding beam propagation and irradiance behavior is essential for safe operation.
At Bahria Town Karachi, KVANT Atom series lasers are used in synchronized fountain installations. This study complements practical experience by applying theoretical modeling and simulation to analyze beam behavior and safety characteristics.

## 2 Theoretical Background

### 2.1 Gaussian Beam Propagation

Laser beams can be approximated as Gaussian beams. The beam radius varies with distance as:

$$w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2} \tag{1}$$

$$z_R = \frac{\pi w_0^2}{\lambda} \tag{2}$$

where $w_0$ is the beam waist radius, $\lambda$ is the wavelength, and $z_R$ is the Rayleigh range.

### 2.2 Irradiance

The peak irradiance is given by:

$$I(z) = \frac{2P}{\pi w(z)^2} \tag{3}$$

where $P$ is the optical power. Irradiance decreases as the beam expands.

### 2.3 Beam Divergence

Beam divergence is expressed as:

$$\theta = \frac{\lambda}{\pi w_0} \tag{4}$$

Lower divergence results in a more collimated beam.

### 2.4 Laser Safety

Laser safety is evaluated using Maximum Permissible Exposure (MPE), which defines safe exposure limits for human eyes and skin.

## 3 Methodology

The simulation uses parameters based on the KVANT Atom 42 laser system.

- Beam diameter at aperture: 7 mm ($w_0 = 3.5$ mm)
- Wavelengths: 638 nm (Red), 525 nm (Green), 455 nm (Blue)
- Power distribution: 9 W (R), 13 W (G), 20 W (B)

Beam propagation and irradiance are computed using Python (NumPy and Matplotlib). The simulation evaluates:

- Beam expansion with distance
- Irradiance decay
- Multi-wavelength (RGB) comparison
- Safety threshold using MPE

# 4 Results and Discussion

## 4.1 Beam Expansion

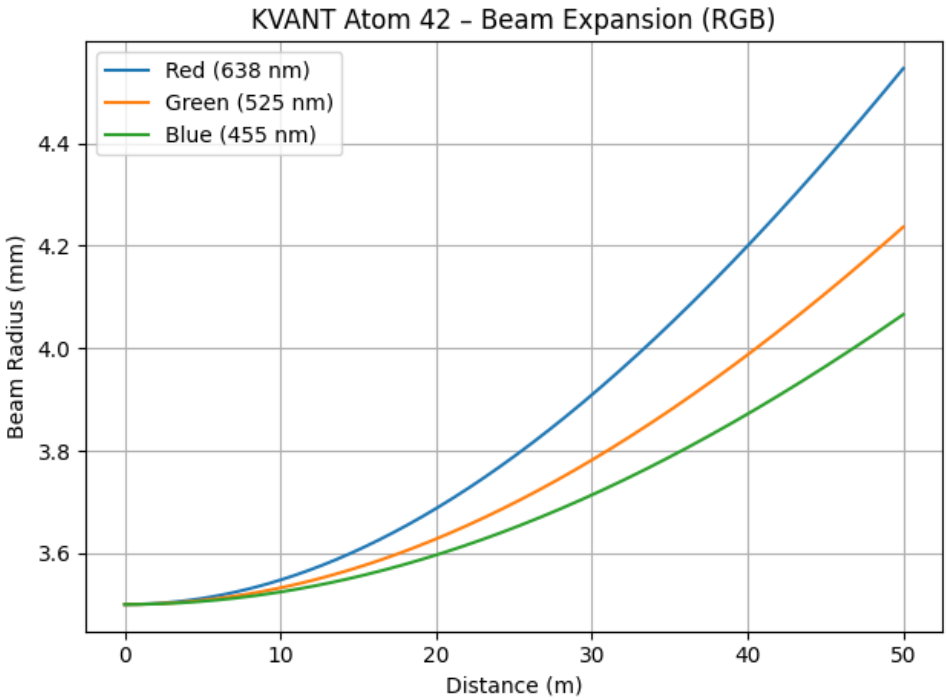


Figure 1: Beam radius variation with distance for RGB wavelengths

The results show gradual beam expansion due to diffraction. Shorter wavelengths exhibit slightly lower divergence.

## 4.2 Irradiance Distribution

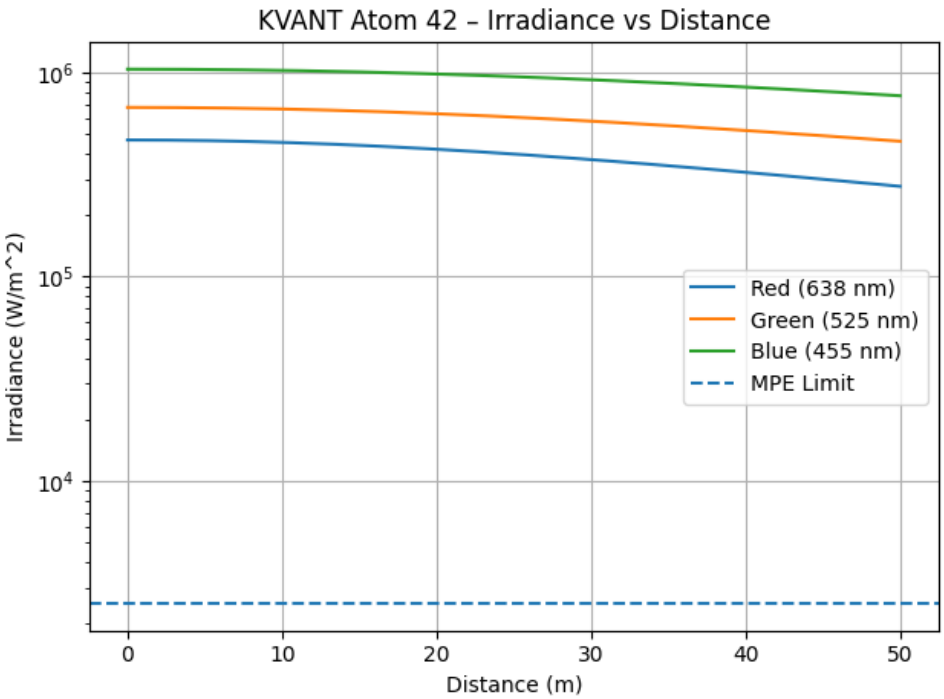


Figure 2: Log-scale irradiance variation with distance

Irradiance decreases rapidly with distance due to beam expansion. Log-scale visualization highlights the wide dynamic range.

### 4.3 Safety Zone Analysis

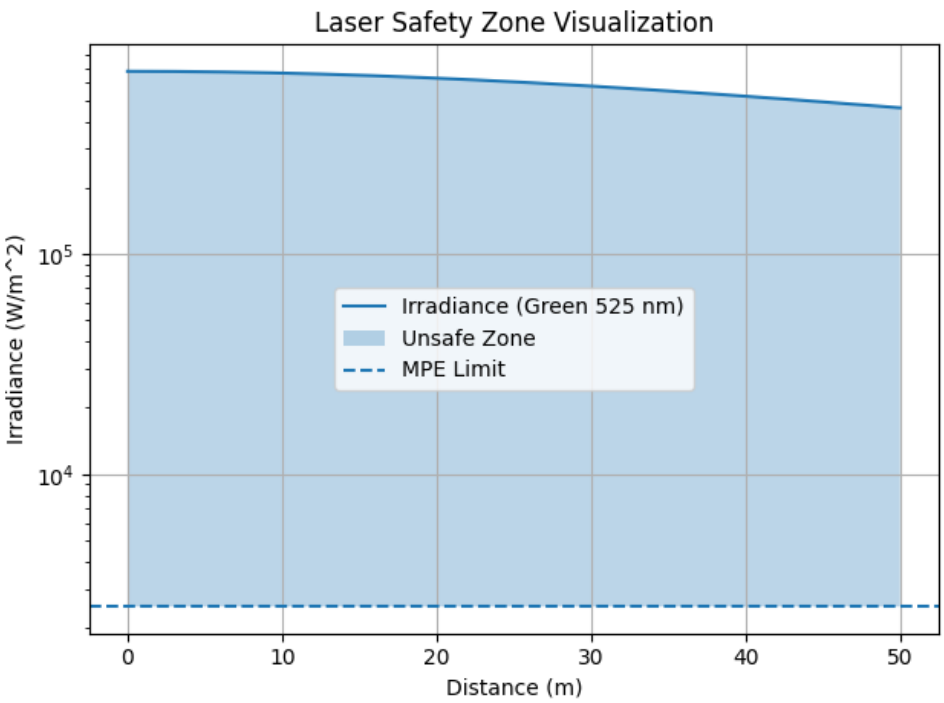


Figure 3: Unsafe regions where irradiance exceeds MPE limits

The shaded region represents distances where irradiance exceeds safe exposure limits.

### 4.4 Discussion

Simulation results indicate very high irradiance near the laser source, which decreases significantly with propagation distance.
The RGB comparison shows that shorter wavelengths maintain higher intensity over longer distances due to reduced divergence. These findings support real-world laser safety practices such as scanning and controlled projection.

## 5 IEC Safety Classification

Laser classification is defined by the IEC 60825 standard. Based on simulated irradiance values exceeding MPE near the source, the KVANT Atom 42 system corresponds to a Class 4 laser.
*Note: This classification is a simplified estimation and does not replace formal IEC evaluation, which requires detailed exposure and system parameters.*

## 6 Practical Relevance

This study supports safe operation of high-power laser systems by providing insight into beam propagation, divergence, and exposure limits. It complements real-world operation of KVANT laser systems in large-scale installations.

# 7 Conclusion

This work demonstrates the application of Gaussian beam theory to real-world laser systems. The results highlight the importance of beam divergence and distance in reducing irradiance. The study provides a foundation for further work in photonics, optoelectronics, and laser safety engineering.

# 8 Future Work

Future extensions may include:

- Time-averaged exposure analysis for scanning beams
- Integration with IEC safety standards
- 3D beam intensity modeling
- Experimental validation using optical instruments

# 9 Tools

Python, NumPy, Matplotlib